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MILITARY SPECIFICATION

PRESSES, ARBOR, HAND-OPERATED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification covers hand operated mechanical presses and hand operated hydraulic presses with equipment and accessories specified herein.

1.2 Classification. This specification covers hand operated arbor presses of the following types, classes, and styles and the sizes shown in tables I through V. The type, class, style and size press required shall be as specified (see 6.2.1).

Type I - Bench or pedestal mounted
  Class 1 - Mechanical
  Class 2 - Hydraulic

Type II - Column or wall mounted, mechanical

Type III - Floor mounted
  Class 1 - Mechanical
    Style A - Elevating table
    Style B - Swinging table

  Class 2 - Hydraulic
    Style A - Standard (non-rolling) table
    Style B - Roller mounted table

2. APPLICABLE DOCUMENTS

2.1 Government documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-TEG, Memphis, Tennessee 38114, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

U. S. DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CFR 1910).

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issue of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/IEEE 268- Metric Practice

(Applications for copies should be addressed to the American National Standard Institute, ATTN: Sales Dept. 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-D-3951 - Commercial Packaging, Standard Practice for.

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

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(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First Article. When specified, (see 6.2.1), a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.2 Design. The press shall be new and one of the manufacturer's current models capable of performing pressing operations in accordance with the requirements herein. The press shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2.1 Measurement systems. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the machine. When only one system of measurements is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all measurements, dimensions, sizes and capacities are given in U.S. units. These measurements may be converted to SI units through the use of the conversion factors and methods specified in ANSI/IEEE 268.

3.2.2 Reclaimed materials. The machine may contain reclaimed materials provided such materials will not jeopardize the machine's intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the machine.

3.2.3 Energy efficiency. The press and all its applicable components that directly consume energy in normal operation shall be designed and constructed for the highest degree of energy efficiency as governed by the latest developments available within the Industry.

3.2.4 Controls. All operating controls shall be located convenient to the operator at his normal work station.

3.2.5 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the press that present safety hazards. The safety devices shall not interfere with the operation of the machine. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance and repair of the parts. All machine parts, components, mechanisms, and

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assemblies furnished on the machine, whether or not specifically required herein, shall comply with all of the requirements of OSHA 2206 that are applicable to the machine itself. Additional safety and health requirements shall be as specified (see 6.2.1).

3.2.6 Mercury restriction. The press shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the press.

3.2.8 Lubrication. Means shall be provided to insure adequate lubrication for all moving parts. When applicable recirculating lubrication systems shall include a filter which is cleanable or replaceable. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings and filler caps shall be accessible.

3.3 Construction. The press shall be constructed of parts which are new, without defects and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free from defects such as scale and mismatching. Processes such as welding, peening, plugging, or filling with solders or pastes shall not be used on castings or forgings to reclaim any defective components for use on the machine. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Welding, brazing or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these processes shall be employed as a repair measure for any defective part.

3.3.3 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked or otherwise permanently installed.

3.3.4 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the press shall be painted in accordance with the manufacturer's commercial practice, provided that all surfaces to be painted shall be cleaned of all foreign matter and not less than two coats of paint are applied.

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3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Gears. All gears shall be constructed from a material suitable for the intended purpose. All gears shall be machined and heat treated by a process that will impart the hardness, surface finish and toughness that will enable the gears to transmit full rated torque without gear damage, failure or premature deterioration and wear. All gears shall be manufactured to definite established standards to provide for interchangeability and parts replacement.

3.4 COMPONENTS

3.4.1 Type I, class I presses.

3.4.1.1 Frame. The type I, class 1 press shall have a "C" or gap type frame. The frame shall support the ram, ram operating mechanism, table and work support plate. The ram shall have accuracy to maintain the ram axis perpendicular to the table surface within 0.010 inch per 6 inches of ram length. The frame table shall have a slot aligned with the axis of the ram that will allow passage of an arbor of the largest diameter specified in table I. The frame throat depth shall accommodate workpieces of the largest diameter specified in table I. The table shall have means for mounting the work support plate (see 3.4.1.2). The table surface supporting the plate shall be finish-machined so that the plate will rest squarely on the table surface without rocking. The frame of type I presses shall be designed for mounting onto a bench, workstand or pedestal. When specified (see 6.2.1), a pedestal or floor stand shall be supplied.

3.4.1.2 Work support plate. All type I presses shall have a circular work support plate having not less than four equally spaced slots. The smallest slot shall be sized to allow passage of an arbor having a diameter 20 to 25 percent of the largest diameter arbor size specified in table I. The other slots shall progressively increase in size, and the largest slot shall be sized to allow passage of an arbor having a diameter 80 to 90 percent of the largest diameter arbor size specified in table I. The plate shall be removable to allow the table slot (see 3.4.1.1) to be used when arbors of the largest diameter specified in table I are to be pressed. The plate shall rotate so that the desired plate slot can be readily positioned in alignment with the ram axis.

3.4.1.3 Ram and ram operating mechanism. The ram shall be extended toward the table by means of a hand lever operated pinion that meshes with a rack machined onto the surface of the ram. Presses having a tonnage rating of not more than 3 tons may have a non-ratcheting type hand lever. Presses having a tonnage rating of more than 3 tons shall have a ratcheting type hand lever and a hand wheel that allows the ram to be rapidly extended to contact the arbor being pressed, and allows the ram to be rapidly retracted after the pressing operation. Ratcheting type hand levers shall have a counterweight, or spring powered mechanism, that raises the lever to the

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TABLE I. Sizes and capacities, type I, class 1 bench or pedestal mounted, mechanical presses.

Size	Rated tonnage	Largest diameter workpiece accommodated, inches	Largest diameter arbor accommodated, inches	Distance, ram to table surface (ram up), inches	Leverage ratio
1	1	7	1	5	25:1
1-1/2	1-1/2	9	1	7-3/4	32:1
2	2	10	1-1/4	8	35:1
3B	3	12	2	11-1/2	45:1
3C	3	12	2	18	45:1
3D	3	14	2-1/4	22	45:1
3E	3	18	3	18	45:1
3F	3	24	2	12	45:1
3G	3	34	2	12	45:1
3H	3	19-1/2	2-1/2	12	60:1
4A	4	15	2-1/2	12-1/2	60:1
4B	4	18	3-5/8	17-1/2	60:1
4C	4	15	2-1/2	18	60:1
4D	4	25	3	12	60:1
4E	4	15	2-1/2	23	60:1
4F	4	25	3	21	60:1
5A	5	18	3-1/2	16	60:1
5B	5	25	3	12	70:1
5C	5	22	5	21	70:1
6A	6	15	2 1/2	15	66:1
6B	6	15-1/2	3	21	66:1
6C	6	15-1/2	5	15	66:1
7	7	27	4-1/2	32	90:1
8	8	18	3-5/8	17-1/2	100:1
10	10	25	3	12	120:1
12A	12	20	5	19	112:1
12B	12	22	3	17	150:1

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uppermost position when hand force is released from the lever. The hand lever may be either of the single lever or compound lever type provided that the leverage ratio (ratio of pounds of ram force to pounds of hand force on the end of the operating lever) conforms to the requirements of table I. Compound lever type handles shall have means for rapidly changing to single-lever type action by moving or sliding a control.

3.4.1.4 Sizes and capacities. The size and capacities of type I, class 1 presses shall conform to the requirements of table I.

3.4.2 Type I, class 2 presses.

3.4.2.1 Frame. Unless otherwise specified, the type I, class 2 press shall be of either the "C" frame or "4" post frame style. When only one style frame is acceptable, the particular style required shall be as specified (see 6.2.1). The frame shall support the ram cylinder and pump assembly. The ram and cylinder axis shall be perpendicular to the work supporting surface of the table within 0.010 inch per 6 inches of ram length. The table structure shall have an opening aligned with the ram axis that will allow passage of an arbor of the largest diameter specified in table II. The worktable shall be designed to accommodate work support devices conforming to 3.4.2.4. The frame configuration shall enable the press to accommodate workpieces up to the largest workpiece diameter specified in table II. The distance between the ram nose and the worktable shall conform to the requirements of table II. The frame shall be designed for mounting onto a bench, workstand or pedestal. When specified (see 6.2.1), a pedestal or floor stand shall be furnished.

3.4.2.2 Hydraulic system. The press shall have a hydraulic system consisting essentially of a fluid reservoir, a pump, a cylinder and ram, a pressure release valve and a pressure gauge. The reservoir shall have a capacity of not less than 1-1/4 times the volume of fluid required to fill the remaining system with the ram at the bottom of its stroke. The design of the pump, cylinder and ram shall enable the press to develop its rated tonnage as required by table II without requiring more than 140 pounds of force applied to the pump lever. The hydraulic system shall have a pressure release valve that, when manually actuated, will release hydraulic pressure back to the fluid reservoir and allow the ram to automatically retract to the top of the stroke position. Means shall be provided to prevent extension of the ram beyond its normal "end-of-stroke" position. The press shall have a pressure gauge calibrated in "tons of ram force" that enables the operator to monitor the ram force being exerted.

3.4.2.3 Ram. The press shall have a means for rapidly extending the ram nose to contact the arbor prior to the arbor pressing operation. The ram extension method shall be either by means of a dual-stage hydraulic pump having a high volume-low pressure stage for rapid ram extension, or by means of a threaded ram extension that can be manually screwed down until it contacts the arbor being pressed. The ram piston travel shall conform to the requirements of table II.

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3.4.2.4 Work support devices. A work support plate and a pair of matched V-blocks shall be furnished with each press. The work support plate shall conform to 3.4.1.2 (substituting "table II" wherever "table I" is mentioned in 3.4.1.2). The matched V-blocks shall be capable of supporting a round bar, or shaft, having a diameter of 1-inch or more, for shaft straightening operations.

3.4.2.5 Sizes and capacities. The sizes and capacities of type I, class 2 presses shall conform to the requirements of table II.

TABLE II. Sizes and capacities, type I, class 2 bench or pedestal mounted, hydraulic presses.

Size	Rated tonnage	Largest diameter workpiece accommodated inches	Largest diameter arbor accommodated inches	Distance ram to table surface, (ram up), inches	Ram piston travel inches	Maximum operating force on pump lever pounds
10	10	17	1-1/4	9-1/2	10	140
10A	10	18	3-1/4	15-3/8	6	140
25	25	13	5-7/8	15	6	140

Requirements are not less than those shown.

3.4.3 Type II presses.

3.4.3.1 Frame. The press shall have a frame conforming to all requirements of 3.4.1.1 (substituting "table III" wherever "table I" is mentioned in 3.4.1.1), except that the frame shall be designed for column or wall mounting in lieu of bench, workstand or pedestal mounting.

3.4.3.2 Work support plate. The press shall have a work support plate conforming to all requirements of 3.4.1.2 (substituting "table III" wherever "table I" is mentioned in 3.4.1.2).

3.4.3.3 Ram and ram operating mechanism. The press shall have a ram and ram operating mechanism conforming to all requirements of 3.4.1.3 substituting "table III" wherever "table I" is mentioned in 3.4.1.3).

3.4.3.4 Sizes and capacities. The sizes and capacities of type II presses shall conform to the requirements of table III.

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TABLE III. Sizes and capacities, type II column or wall mounted, mechanical presses.

Size	Rated tonnage	Largest diameter workpiece accommodated, inches	Largest diameter arbor accommodated, inches	Distance ram to table surface, (ram up), inches	Leverage ratio
3	3	.14	.2	11-1/4	48:1

Requirements are not less than those shown.

3.4.4 Type III, class 1 presses.

3.4.4.1 Frame. The type III, class 1 press shall have a "C" or gap type frame and be designed for floor mounting. The frame shall support one or more worktables, a ram and a ram operating mechanism. The ram shall have sufficient accuracy to maintain the ram axis perpendicular to the table surface within 0.010 inch per 6 inches of ram length. The type III, class 1, style A worktable shall conform to 3.4.4.2.1 and the type III, class 1, style B worktable shall conform to 3.4.4.2.2.

3.4.4.2 Worktable.

3.4.4.2.1 Style A. The type III, class 1, style A press shall have a worktable which can be elevated or lowered on a way or slide machined into the face or column of the press frame. The slide shall have a gear rack to allow the table to be raised or lowered by means of a hand cranked pinion mechanism. The press shall have a mechanism that securely clamps the table to the column at any position within the table movement range. The worktable working surface shall be machined and shall have provisions for mounting a removable work support plate conforming to 3.4.4.3 and a slot that will allow passage of an arbor of the diameter specified in table IV. The distance from the ram to the worktable surface, with the ram raised and table lowered, shall conform to the requirements of table IV.

3.4.4.2.2 Style B. The type III, class 1, style B press shall have sturdy pivots to support two swinging tables, one above the other and allow both tables, individually, to swing into and out of normal operating position. Means shall be provided for locking the worktables in the normal operating position. The upper table surface, lower table surface and base of the press shall be usable to support the workpiece during arbor pressing operations. Each table and the base shall have a machined working surface and provisions for mounting a removable work support plate conforming to 3.4.4.3. Each swinging worktable and the base shall have a slot aligned with the ram axis that will allow passage of an arbor of the diameter

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specified in table IV. The distance from the ram to the table surfaces and base work surfaces shall conform to the requirements of table V.

3.4.4.3 Work support plate. Type III, class 1, style A and style B presses shall have a work support plate conforming to all requirements of 3.4.1.2 (substituting "table IV" for style A presses and table V for style B presses wherever "table I" is mentioned in 3.4.1.2).

3.4.4.4 Ram and ram operating mechanism. All type III, class 1, style A and style B presses shall have a ram and ram operating mechanism conforming to all requirements of 3.4.1.3 (substituting "table IV" for style A presses and table V for style B presses wherever "table I" is mentioned in 3.4.1.3).

3.4.4.5 Arbor catcher (type III, class 1, style A presses). Type III, class 1, style A presses shall have a device that catches the falling arbor as it is pressed out of the workpiece. The device shall consist essentially of a cylindrical or box shaped container lined with a soft metal, such as babbitt, copper, or other suitable metal alloy. The device is not required to have moving parts since it is intended only to "arrest-the-fall" of heavy arbors as they are pressed out of the workpiece, preventing damage to the arbor or the machine base. The device shall be of adequate size to accommodate arbors of the largest diameter specified in table IV and shall be securely mounted on the machine base in alignment with the ram axis.

3.4.4.6 Sizes and capacities. The sizes and capacities of the type III, class 1, style A presses shall conform to the requirements of table IV and the size and capacity of the type III, class 1, style B presses shall conform to the requirements of table V.

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TABLE IV. Sizes and capacities, type III, class 1, style A floor mounted, mechanical, elevating table presses.

Size	Rated tonnage	Largest diameter workpiece accommodated, inches	Largest diameter arbor accommodated, inches	Distance ram to table surface (ram up), inches	Leverage ratio
5	5	20	4-1/2	26	55:1
7	7	27	4-1/2	26	90:1
12	12	10	4-1/2	26	112:1
12A	12	21	4-1/2	30	160:1
15	15	27	4-1/2	32	200:1

Requirements are not less than those shown.

TABLE V. Sizes and capacities, type III, class 1, style B floor mounted, mechanical, swinging table presses.

Size	Rated tonnage	Largest diameter workpiece accommodated, inches	Largest diameter arbor accommodated, inches	Distance, ram to work support plates and base work surface (ram up), inches			Leverage Ratio
				Top table plate	Lower table plate	Base	
15	15	24	5-1/2	16	30	46	150
25	25	37	5-1/2	16-1/2	31-1/2	47	220:1

Requirements are not less than those shown.

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3.4.5 Type III, class 2 presses.

3.4.5.1 Frame. The type III, class 2 press shall have an "H" type frame. The frame shall be fabricated of structural sections such as I-beams, channels, angles, bars, plates, or any combination thereof. The side members shall be rigidly joined together by three or more crossmembers. One crossmember shall serve as a headrail that supports the hydraulic workhead. The crossmember directly below the headrail shall serve as the worktable, and one or more additional crossmembers shall provide additional rigidity to the frame and maintain parallelism of the side members as the table or headrail, as applicable, is raised or lowered. The frame shall have feet of adequate length and strength to stabilize the frame structure and prevent it from tilting or tipping during press operation. In lieu of the crossmember type table described above, type III, class 2, style B presses shall have a track that supports a roller mounted table that can be rolled out of normal operating position to facilitate loading of heavy workpieces onto the table by means of an overhead crane or hoist. The frame structure, table structure, table support mechanism, and workhead of all type III, class 2 presses shall have sufficient accuracy and rigidity to maintain the ram axis perpendicular to the working surface of the table to an accuracy of 0.050 inch per foot.

3.4.5.1.1 Frame siderails. The siderails shall be either of the open or closed type. The siderails shall have a series of equally spaced holes that allow either the headrail or worktable, or both, to be raised or lowered. The hole spacing shall provide a maximum distance between the ram nose and the worktable in accordance with the requirements of table VI. The siderails shall support a manually operated, self-locking winch. A set of hardened steel pins shall be provided for supporting the headrail or worktable to the siderails. The pins shall be shaped for easy insertion and removal from the holes. The total shear strength of the pin set shall be not less than double the tonnage rating of the press. The table and table support pins shall be capable of withstanding the bending stresses encountered when operating at 110% of the rated press tonnage without permanent deformation of the table or table support pins. The pin hole spacing in all members shall be to the precision necessary to insure that all pins support equal loads when the workhead is centered on the headrail. The width of the press between the frame siderails shall conform to the requirements of table VI. The type III, class 2, style B press shall be furnished with a manually operated, self-locking winch with necessary cables, chains and pulleys for raising and lowering the headrail or table, as applicable. When specified (see 6.2.1) this type press shall be furnished with a power driven threaded lift. The lift shall have means to operate either mechanically or with a standard 1/2 inch drive electric drill. When specified (see 6.2.1) the 1/2 inch electric drill shall be furnished.

3.4.5.1.2 Worktable. The worktable shall be designed to directly support the workpiece, or to support workholding devices that support the workpiece (see 3.4.5.6). The width between table channels on style A

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presses shall conform to the requirements of table VI. The width between table channels on style B presses shall be adjustable over the range specified in table VII. The worktable of style A machines shall be of the type that is permanently located below the ram axis under all conditions of operation. The worktable for style B presses shall be mounted on rollers that roll in a track mounted on the press frame. The rolling worktable shall be easily rolled into and out of normal operating position below the ram axis so that large workpieces can be readily loaded onto the table in the normal operating position. The table travel distance shall be such that when the table channels are adjusted to the spacing specified in table VII, and when the table is in the "loading" position, all table channel edges shall be beyond a perpendicular dropped from the side of the headrail structure on the side of the headrail closest to the extended worktable.

3.4.5.2 Hydraulic workhead. Unless otherwise specified, the hydraulic workhead shall be the fixed type. When specified the hydraulic workhead shall be the rolling type (see 6.2.1). Fixed workheads shall be permanently and securely attached to the center of the headrail. Rolling type workheads shall have means for secure clamping at any adjusted position within the range of movement. Presses having a rolling type workhead shall be capable of performing full rated tonnage pressing operations with the workhead at any position within its range of travel without deformation of any component or structural member. The workhead shall contain the hydraulic cylinder, ram piston, ram and ram extension mechanism. The hydraulic pump may be mounted directly on the workhead, or may be mounted at any convenient operating position on the press frame.

3.4.5.3 Hydraulic system. The hydraulic system shall consist essentially of a fluid reservoir, hydraulic pump, cylinder and ram assembly, pressure release valve, pressure gauge and necessary piping, hoses and fittings. The hydraulic system design shall enable the press to deliver the rated tonnage, specified in table VI or VII, with not more than 140 pounds of force applied to the end of the pump lever. All components in the hydraulic system shall have a safe working pressure rating not less than 120% of the pressure that can be exerted by the pump. Means shall be provided to prevent loading the press beyond 110% of its rated tonnage. Means shall also be provided to prevent extension of the ram beyond its normal "end-of-stroke" position. All components in the system shall be sealed to prevent fluid leakage when the system is operating at full tonnage capacity.

3.4.5.3.1 Fluid reservoir. The fluid reservoir shall have adequate capacity to hold not less than 1-1/4 times the volume of fluid required to fill the remaining system with the ram at the bottom of its stroke.

3.4.5.3.2 Hydraulic pump, cylinder and ram. Unless otherwise specified, the press shall be furnished with one of the types of pump, cylinder and ram mechanism described in 3.4.5.3.2.1 through 3.4.5.3.2.3. When only one type pump cylinder and ram mechanism is acceptable, the particular type required shall be as specified (see 6.2.1). Regardless of the system

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furnished, the pump shall have a lever designed to insure comfortable hand operation and to prevent lever distortion when the press is developing maximum rated capacity. The pump, or some other part of the hydraulic circuit, shall have a pressure release valve that will release pressure back to the reservoir allowing the ram to be returned to the "top- of-stroke" position.

3.4.5.3.2.1 Dual-stage pump type system. This type system shall consist of a pump that has high volume-low pressure stage and a low volume-high pressure stage. The high volume-low pressure stage shall provide rapid ram extension to the arbor prior to the arbor pressing operation, and, when used with a double acting ram cylinder having no spring return mechanism, shall provide rapid return of the ram to the "top-of-stroke" position. The low volume-high pressure stage shall be capable of delivering adequate pressure to enable the press to exert the rated tonnage specified in table VI and VII, without requiring more than 140 pounds of force on the end of the pump operating lever.

3.4.5.3.2.2 Single-stage pump with rack and pinion ram extension mechanism. This type system shall consist of a single pump capable of developing adequate pressure to enable the press to exert the tonnage specified in table VI or VII, without requiring more than 140 pounds of force on the end of the pump operating lever. The ram shall have a mechanical system, such as a handwheel operated rack and pinion mechanism, that provides rapid ram extension to the arbor prior to the arbor pressing operation and returns the ram to the "top-of-stroke" position after completion of the pressing operation and after actuation of the pressure release valve.

3.4.5.3.2.3 Single-stage pump with threaded ram extension mechanism. This type system shall consist of a single-stage pump capable of developing adequate pressure to enable the press to exert rated tonnage specified in table VI or VII, without requiring more than 140 pounds of force on the end of the pump operating lever. The ram shall have a threaded extension for rapidly extending the ram to contact the arbor prior to the actural arbor pressing operation.

3.4.5.4. Auxiliary 3 ton, rapid ram movement capabilities. Unless otherwise specified (see 6.2.1), all type III, class 2 presses are required to have means of exerting a ram force of not less than 3-tons without requiring a large number of pump strokes. This may be accomplished by:

(a) Utilizing a dual speed pump system (see 3.4.5.3.2.1) capable of providing not less than 3-ton ram force.

(b) Utilizing a single-stage pump with a rack and pinion type ram extension device (see 3.4.5.3.2.2) capable of mechanically exerting not less than 3-ton force at a leverage ratio of not less than 48 to 1. If the pump system does not meet the requirements of (a) or (b) above, an auxiliary 3-ton mechanical press meeting or exceeding the requirements of type I for

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class 1, size 3B, of table I, or type II, size 3 of table III and the requirements of this specification shall be furnished. The 3-ton auxiliary mechanical press, when provided, shall be securely mounted at a convenient location on the press frame.

3.4.5.5 Pressure gauge. All hydraulic presses shall have a pressure gauge calibrated in tons of ram force. The gauge shall be legible from the operators normal work station and shall not be obstructed by any part of the press.

3.4.5.6 Work support device. All type III, class 2 presses shall have two work support plates that rest on the table channels and support the workpiece. The work support plates shall be movable along the entire width of the table to provide the necessary support for any workpiece within the size range of the machine. The upper surface of the plates shall be flat and the bottom surface shall have a tongue, step, shoulder, or other means that will hold the plate in alignment, preventing the plate from sliding off the table, but allow free sliding motion on the table surface. In addition, a pair of V-block type work support plates, suitable for supporting a round bar during straightening operations shall be furnished with each press. If the supplier normally furnished a single pair of invertable blocks suitable for supporting flat surfaces using one side, and supporting round bars using the other side, the pair of invertable blocks may be supplied in lieu of the two pairs of supports required above.

3.4.5.7 Sizes and capacities. The sizes and capacities of type III, class 2 style A presses shall conform to the requirements of table VI. The size and capacities of the type III, class 2, style B presses shall conform to the requirements of table VII.

3.5 Performance requirements. All presses covered by this specification shall be capable of performing any function for which they are designed and within their rated tonnage capacity. The presses covered herein shall be capable of exerting a force equal to 110 percent of their maximum rated tonnage capacity without permanent distortion or other damage to any component.

3.5.1 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2.1).

3.6 Alignment tolerances. Each press shall meet the alignment accuracies specified in paragraphs 3.4.1.1, 3.4.2.1, 3.4.4.1 and 3.4.5.1.

3.7 Standard equipment. Unless otherwise specified (see 6,2.1), the presses shall be furnished with all standard equipment normally furnished by the manufacturer including the following:

(a) One set of tools normally furnished with each press.

(b) Two ram noses; one flat, and one vee.

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TABLE VI. Sizes and capacities, type III, class 2, style A floor mounted, hydraulic , non-rolling table presses.

Size	Rated tonnage	Width between frame siderails, inches	Width between table channel maximum arbor size, inches	Distance ram nose to table, inches	Vertical travel of table or headrail, inches	Ram piston travel, inches
10	10	18	3	20	15	2
17	17	20	5	34	35	2
20	20	24	5	37	34	5
25	25	24	5	39	35	5
30	30	28	6	32	35	5
40	40	28	7	40	36	5
50	50	28	7	40	35	4
60	60	35	8	40	36	7
75	75	35	8	36	30	4-1/2
80	80	45	8	36	30	7
100	100	45	9	36	30	8
100A	100	28	9	36	30	6
125	125	48	9	25	20	8
150	150	48	10	25	20	8
200	200	48	15	30	24	13

Requirements are not less than those shown.

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TABLE VII. Sizes and capacities, type III, class 2, style B floor mounted, hydraulic, roller mounted table presses.

Size	Rated tonnage	Width between upright columns, inches	Width between table channels adjustable over a range of, inches	Table channel spacing for unobstructed table loading, inches 1/	Distance ram nose to table inches 2/	Vertical travel of table or headrail, inches	Ram piston travel, inches
50	50	42	0 to 29	8	39	28	11
75	75	44	0 to 29	10	39	28	11
80	80	48	4 to 27	14	60	48	13
100	100	48	4 to 27	11	60	44	13
150	150	48	4 to 27	9	58	44	13
200	200	51	4 to 27	9	63	55	13

1/ These dimensions represent the widest table channel spacing that will allow the table to be rolled to a position in which all table channel edges are beyond a perpendicular dropped from the headrail structure (see 3.4.5.1.2).

2/ This is the distance between the ram nose and the table. This distance is with the ram nose retracted and measurement taken from the flat of the ram nose to the table.

Requirements are not less than those shown. Where a range is shown, the required performance is from the smaller figure, or less, to the larger figure, or more.

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3.8 Optional equipment. Optional equipment shall be furnished as specified and shall be fully described (see 6.2.1).

3.9 Repair parts. Such repair parts as are specified (see 6.2.1), shall be furnished.

3.10 Marking of instruction plates, indicating plates and name plates. All words on instruction and indicating plates shall be in the English language. Characters shall be engraved, etched, embossed or stamped in boldface on a contrasting background.

3.10.1 Lubrication chart or plate. Unless otherwise specified (see 6.2.1), a lubrication chart or plate shall be permanently and securely attached to each press. If a chart is furnished, it shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets, with suitable means of mounting. The chart or plate shall contain the following information:

Points of lubricant application.
Servicing interval
Type of lubricant
Viscosity

3.10.2 Nameplate. Unless otherwise specified (see 6.2.1), a nameplate shall be securely attached to each press. The nameplate shall contain the information listed below. If the press is a special model, the model designation shall include the model of the basic standard press and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.11 Technical data. When technical data is required it shall be furnished in accordance with the requirements of the contract (see 6.2.2).

3.12 Workmanship. Workmanship of the press and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise

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specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.5 and all tests in 4.6. Failure of the item to pass the first article tests shall be cause for rejection.

4.4 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1) quality conformance inspection shall consist of at least the examination in 4.5 and the tests in 4.6.3. Failure of the item to pass the examination or tests shall be cause for rejection.

4.5 Examination. The press and equipment shall be examined to determine compliance with the requirements of 3.2 through 3.5 and 3.7 through 3.12.

4.6 Tests

4.6.1 Operational test. The press shall be operated at no load for not less than 30 minutes to assure proper operation of all controls, adjusting mechanisms, and accessories.

4.6.2 Alignment accuracy tests. Each press shall be tested to determine that it complies with the alignment accuracy requirements of 3.6.

4.6.3 Performance tests. Each press shall be loaded and tested for proper operation when loaded to 110 percent of maximum rated capacity. Type I, class 2 and type III, class 2 presses shall retain the load for a period of not less than one hour. Either during or after testing, the press shall be examined for permanent distortion, breakage, hydraulic leaks, proper operation of the pressure gauge and ram retraction mechanism, and any other evidence of nonconformance to this specification.

4.7 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of Section 5.

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5. PACKAGING

5.1 Packaging, packing and marking. Unless otherwise specified, packaging, packing, and marking shall be in accordance with ASTM-D-3951. When specified (see 6.2.1), level A preservation level A or level B packing and special marking shall be accomplished in accordance with MIL-M-18058.

6. NOTES

6.1 Intended use. The presses covered by this specification are for use in machine, production, and repair shops at shore facilities and on ships.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type, class, style and size required (see 1.2).

c. When first article is required for inspection and approval (See 3.1).

d. If machine is required to be configured in a specific measurement system (US or SI), state required system (see 3.2.1).

e. Additional safety and health requirements, if required (see 3.2.5).

f. Painting, if different (see 3.3.5).

g. Pedestal or floor stand, if required (see 3.4.1.1).

h. If press frame style is to be a specific type, state required style (see 3.4.2.1).

i. Pedestal or floor stand, if required (see 3.4.2.1).

j. Power driven threaded lift, if required (see 3.4.5.1.1).

k. Electric drill, if required (see 3.4.5.1.1).

l. Hydraulic workhead, if different (see 3.4.5.2).

m. If hydraulic pump, cylinder and ram mechanism is to be a specific type, state required type (see 3.4.5.3.2).

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n. Auxiliary high speed 3 ton pressing capability, if different (see 3.4.5.4).

o. Additional performance features, if required, (see 3.5.1).

p. Standard equipment, if different (see 3.7).

q. Specify and fully describe optional equipment required, if different (see 3.8).

r. Specify applicable repair parts, if required (see 3.9).

s. Lubrication chart or plate, if different (see 3.10.1).

t. Nameplate, if different (see 3.10.2).

u. First article inspection, if different (see 4.3).

v. Quality conformance inspection, if different (see 4.4).

w. Level of packaging, packing and marking, if different (See 5.1).

6.2.2 Contract data requirements. Required technical data such as manuals, parts lists, wiring diagrams, and examination, original operators and tolerance test reports and other instructions for operation alignment and maintenance (as identified on a DD Form 1664, Data Item Description) should be specified on a DD Form 1423, Contract Data Requirements List incorporated into the contract.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test and approval of the first article.

6.4 Safety and health requirements. Paragraph 3.2.5 requires compliance only with those OSHA requirements that concern the machine itself. It does not require compliance with those OSHA requirements that concern "the machine in its operating environment" such as noise levels, radiation levels electromagnetic emissions, noxious vapors, air contaminants, heat, etc. Since OSHA limits the total hazard level of these hazards in the environment and does not limit the hazard level of individual machines in the environment the procuring activity is advised to analyze the existing hazard levels in the proposed operating environment, and specify additional machine requirements that will integrate the new machine into its future operating environment. The machine shall be equipped with all points-of-operation guarding normally furnished as standard on the manufacturer's commercial machines supplied to the commercial market. If specific point-of-operation guarding is required, the procuring activity should specify the exact

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configuration of the guard required, as in most cases, the guard configuration is dependent on the size and configuration of the workpieces. The above, and any other additional safety and health requirements should be specified in detail under 6.2.1 (e).

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - AL
Navy - SH
Air Force - 99

Review Activities:
Navy - YD
DLA- GS

User Activities:
Army - AV, ME, WV
Navy - AS, MC

Preparing Activity:
DLA-IP

Project Number:
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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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